Software Requirements Specification

for

WalkingSG

Version 5.0

Prepared by Trinh Hoang Huy

Group 3 – SCS2

September 5th

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Table of Contents

Table of Contents ..... **ii**
Revision History ..... **iii**
1. Introduction **4**
1.1 Purpose 4
1.2 Document Conventions 4
1.3 Intended Audience and Reading Suggestions 4
1.4 Product Scope 4
1.5 References 4
2. Overall Description **5**
2.1 Product Perspective 5
2.2 Product Functions 5
2.3 User Classes and Characteristics 5
2.4 Operating Environment 5
2.5 Design and Implementation Constraints 5
2.6 User Documentation 5
2.7 Assumptions and Dependencies 5
3. External Interface Requirements **6**
3.1 User Interfaces 6
3.2 Hardware Interfaces 11
3.3 Software Interfaces 12
3.4 Communications Interfaces 12
4. System Features **12**
4.1 Authentication 12
4.2 Preference Management 13
4.3 Search & Map Interface 14
4.4 Routing & Scoring Engine 15
4.5 Live Navigation 17
4.6 Community Hazard & Local Feed 17
5. Other Nonfunctional Requirements **18**
5.1 Performance Requirements 18
5.2 Safety Requirements 18
5.3 Security Requirements 19
5.4 Software Quality Attributes 19
5.5 Business Rules 19
6. Other Requirements **20**
Appendix A: Analysis Models **20**
Appendix B: To Be Determined List **24**
Appendix C: Data Dictionary **24**
Appendix D: Use case description **27**
6.1 Use Case Descriptions 27

Revision History

Name	Date	Reason For Changes	Version
Trinh Hoang Huy	August 31st	Initial Requirements Proposal	1.0
Hemanth	September 4th	Update Data Dictionary	2.0
Trinh Hoang Huy	September 5th	Refine Functional Requirements	3.0
Lin Ziqi	September 5th	Update UI Update Content page	4.0
Trinh Hoang Huy	September 5th	Final touch on the first SRS Version (Reformat for better look)	5.0
Fujita Natsuo	September 16th	Refined UI	6.0
Trinh Hoang Huy	September 17th	Fill in the remaining placeholders	7.0
Lin Ziqi	September 18th	Added in class diagram of entity class	8.0
Fujita Natsuo	September 18th	Added in initial dialog diagram	9.0
Benjamin	October 25th	Added use case description, updated appendix A	10.0

1. Introduction

1.1 Purpose

- This system is a pedestrian-first, mobile-based navigation application scoped specifically to the geographic boundaries of Singapore. It provides personalized pedestrian routing based on user mobility constraints, real-time weather, and community-driven accessibility scoring.

1.2 Document Conventions

- Functional requirements are identified by a unique sequence tag (e.g., REQ-AUTH-01). Priorities (High, Medium, Low) are explicitly stated for major features and inherited by their sub-requirements.

1.3 Intended Audience and Reading Suggestions

- This document is intended for backend developers, frontend mobile developers, UI/UX designers, and project managers. Developers should focus heavily on Section 4 (System Features) and Section 5 (Other Nonfunctional Requirements), while designers should focus on Section 3 (External Interface Requirements).

1.4 Product Scope

- WalkingSG aims to solve the "last mile" and pedestrian navigation gaps left by vehicle-centric apps. The software allows users to define physical mobility needs (e.g., wheelchair) and environmental preferences (e.g., maximum shelter). It calculates five distinct routes for every query, ranking the top choice dynamically based on the user's profile. It relies on community reports to adjust the "confidence score" of individual map segments, ensuring real-time hazard avoidance.

1.5 References

- Singapore Land Authority (SLA) OneMap API Documentation (v2.0 RESTful Services: Search, Reverse Geocoding, Barrier-Free Pedestrian Routing).
- Singapore Active Mobility Act 2017 (Cap. 2C) (Pedestrian and mobility aid regulations).
- Singapore Land Transport Authority (LTA) Walk2Ride Programme & Code of Practice for Street Works (Sheltered Walkways & Accessibility Guidelines).
- National Environment Agency (NEA) & Meteorological Service Singapore (MSS) Weather and UV Index Data APIs.
- React Native & Expo Documentation (Expo SDK 52+, React Navigation v6, React Native 0.86).

2. Overall Description

2.1 Product Perspective

- WalkingSG is a new, self-contained mobile application. It operates as a front-end interface that relies heavily on external mapping APIs (OneMap, OpenStreetMap) for baseline geospatial data, while maintaining its own proprietary database for user profiles, saved preferences, and custom map-segment confidence scores.

2.2 Product Functions

- Account creation and password recovery.
- Management of personal mobility and environmental preferences.
- Calculation and display of 5 distinct pedestrian route types (Direct, Maximum Shelter, Barrier-Free, Low-Effort, Wide/Quiet).
- Live, GPS-based turn-by-turn navigation.
- Community hazard reporting and dynamic path-segment scoring.
- Location-based Local Hazard News Feed.

2.3 User Classes and Characteristics

- **Standard Pedestrians:** High frequency of use, seeking sheltered or direct routes.
- **Mobility-Impaired Users (Wheelchair):** Require high accuracy on step-free ("Barrier-Free") routing. Zero tolerance for unmapped stairs.
- **Elderly Walkers:** Prefer "Low-Effort" routes avoiding steep inclines.

2.4 Operating Environment

- The software will operate as a cross-platform mobile application (iOS 15+ and Android 10+). It requires an active internet connection (4G/5G/WiFi) and active device GPS functionality.

2.5 Design and Implementation Constraints

- Geographic scope is strictly locked to Singapore's coordinate bounding box.
- The system shall not calculate vehicular, bicycle, or public transit routing. It is exclusively optimized for walking and pedestrian-supportive personal mobility aids.
- Data consumption must be minimized to maintain performance on typical mobile data plans.

2.6 User Documentation

- **Interactive In-App Onboarding Flow**: First-launch tutorial explaining GPS permissions, Singapore bounding geofence, and Personal Declaration setup.
- **WalkingSG User Manual (PDF / Markdown)**: End-user instructions for route variant selection (Direct, Sheltered, Barrier-Free, Low-Effort, Wide/Quiet), live navigation, and community reports.

2.7 Assumptions and Dependencies

2.7.1 Assumptions

- **GPS Signal Availability:** It is assumed that the end-user's device has a functional GPS sensor capable of maintaining an accuracy threshold of within 10 meters, and that Singapore's urban canyons (e.g., the Central Business District or Marina Bay) do not consistently block satellite positioning signals.
- **Continuous Cellular Internet:** It is assumed that users maintain active mobile data packages (4G/5G) or are connected to urban public WiFi networks during routing queries and live navigation.
- **Active Mobility Compliance:** It is assumed that standard pedestrians and mobility-impaired users comply with physical path markers, speed regulations under the Singapore Active Mobility Act, and real-world construction signs, treating app recommendations as advisory.

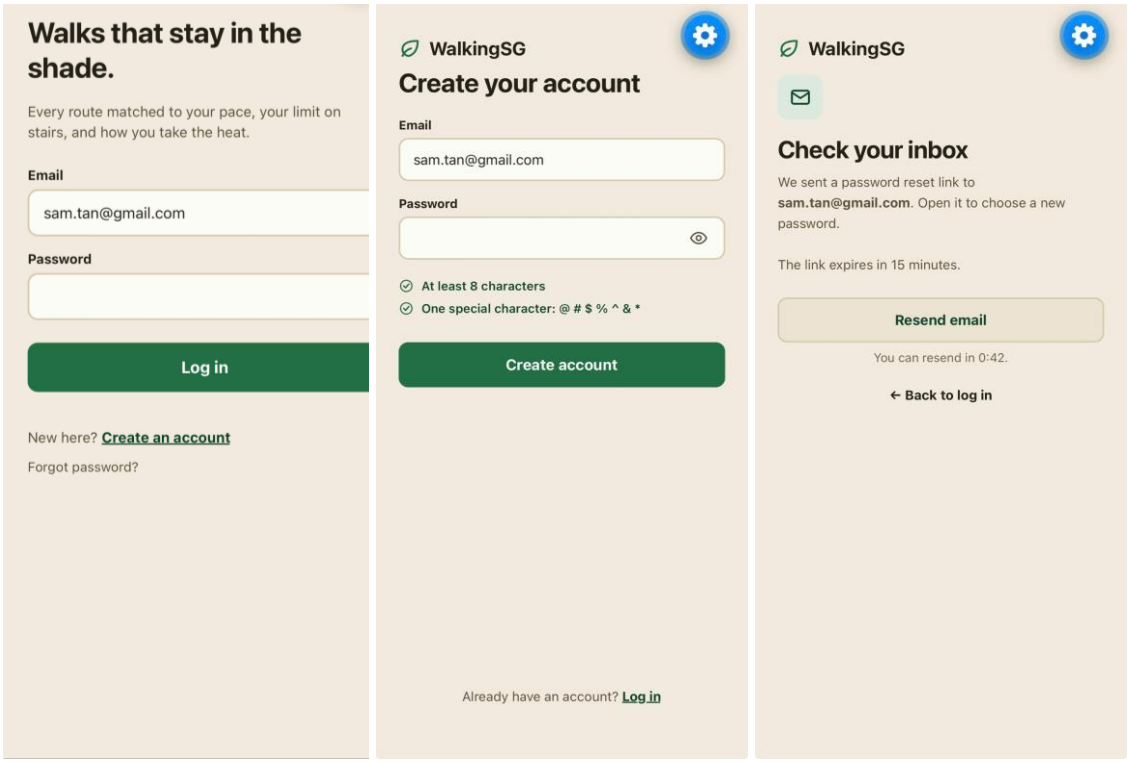
2.7.2 Dependencies

- **Singapore Land Authority (SLA) OneMap APIs:** The system depends entirely on SLA's OneMap APIs for baseline address lookup, autocomplete suggestions, base map tiles, and foundational wheelchair-accessible networks. Any service disruption on SLA's production server directly degrades routing capabilities.
- **Firebase Cloud Messaging (FCM) & Apple Push Notifications (APNs):** The instant local hazard news broadcasting mechanism relies on steady integrations with third-party cloud messaging push pipelines to reach active devices within 5-kilometer bounding boxes.

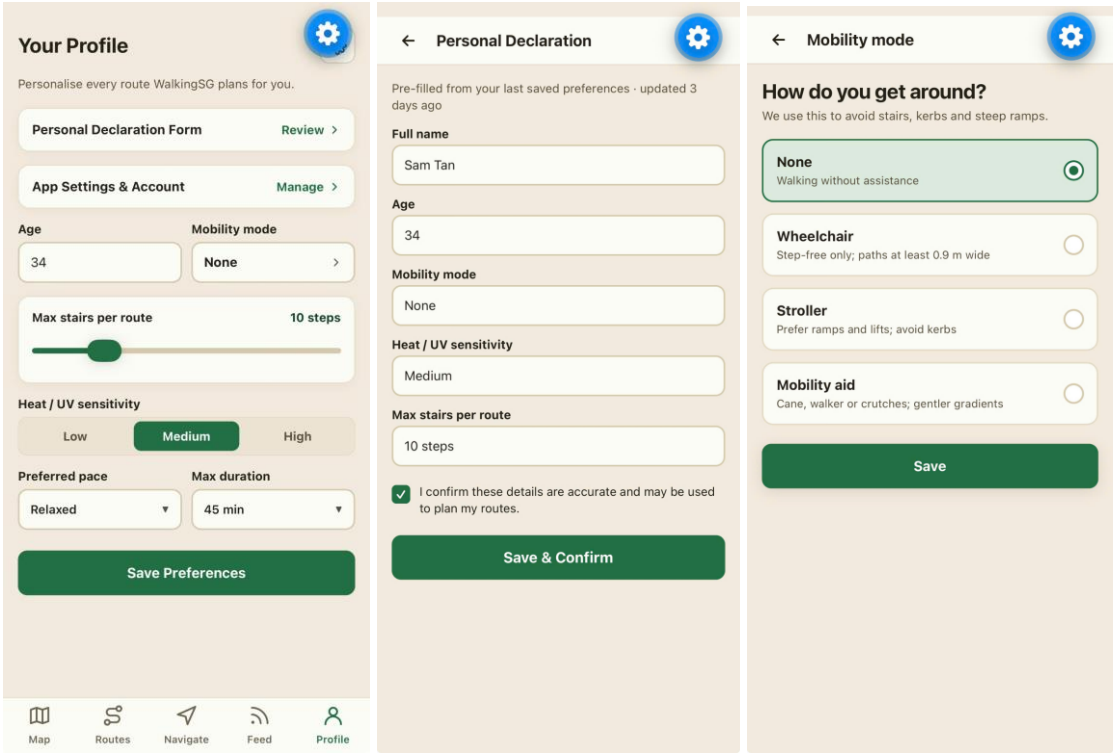
3. External Interface Requirements

3.1 User Interfaces

Login Screen



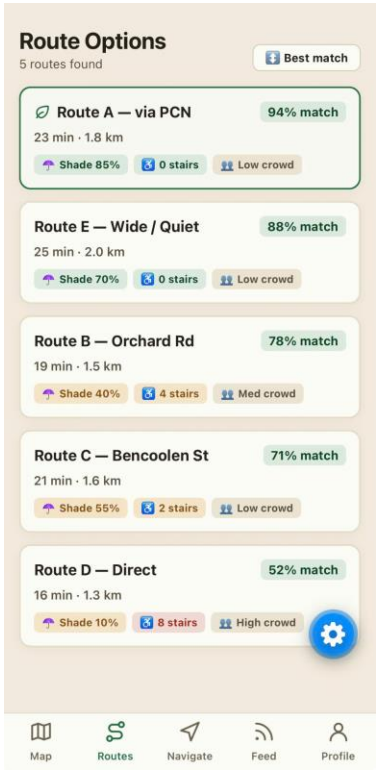
Profile and Preferences Screens



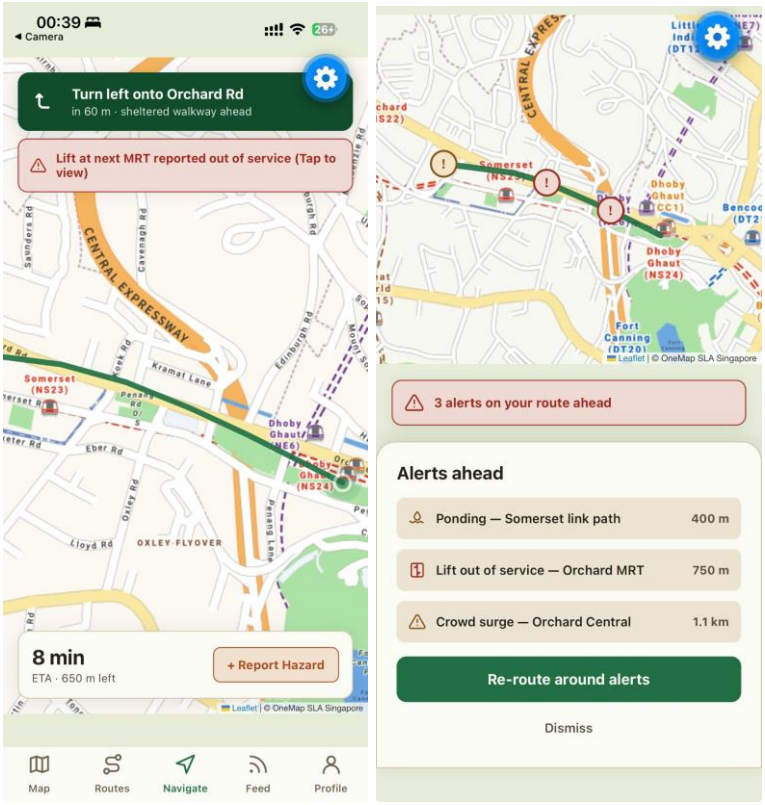
Search and Map Screen



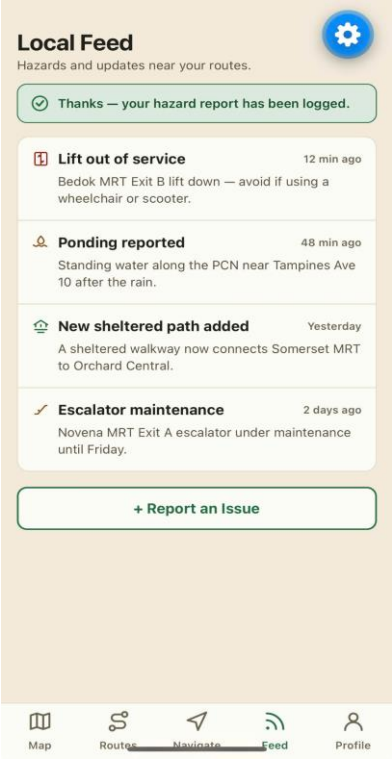
Route Result Screen

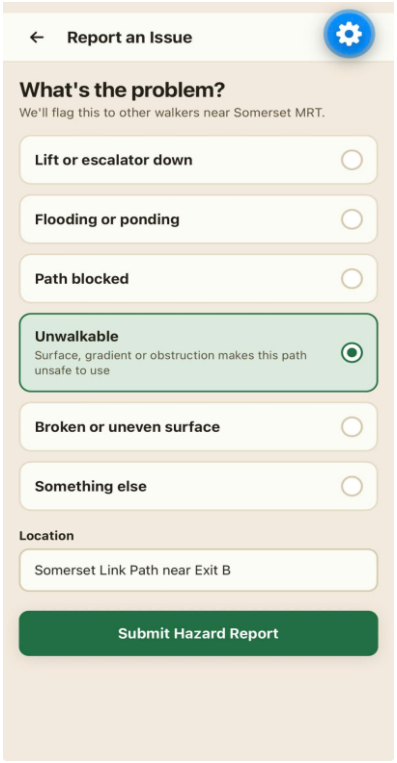


Live Navigation Screen

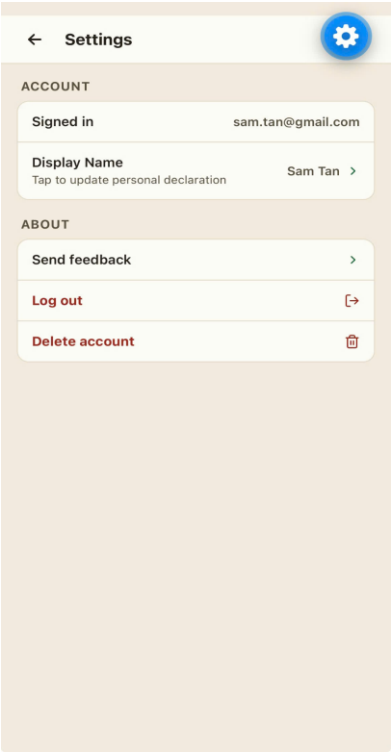


Local Feed Screen





Setting Screen



3.2 Hardware Interfaces

- The software must interface with the mobile device's internal GPS sensor to receive continuous coordinate updates (latitude/longitude/heading) during active navigation.

3.3 Software Interfaces

- **OneMap API:** Used for address autocomplete, base map rendering, and baseline barrier-free routing data.
- **Database (e.g., PostgreSQL):** Stores user profiles, hashed passwords, and path-segment confidence tables.

3.4 Communications Interfaces

- **Client-Server Protocol:** HTTPS for all REST API endpoints**.**
- **Mapping & Geolocation Services:** External SLA OneMap HTTPS REST API for address search, geocoding, and polyline coordinates.
- **Transactional Email Dispatch:** Secure SMTP / RESTful email service (TLS encrypted) for 6-digit registration OTPs and password reset tokens.
- **Network Resiliency:** 5.0s client timeout threshold with exponential backoff retry mechanism (max 3 attempts) for cellular connectivity.

4. System Features

4.1 Authentication

4.1.1 Description and Priority

- **High priority**. Handles secure user access, account creation, credential validation, and session lifecycle management.

4.1.2 Stimulus/Response Sequences

- User opens app -> System prompts login/register -> User submits credentials -> System validates against database -> System authenticates, generates session token, and grants access.

4.1.3 Functional Requirements

REQ-AUTH-01: The system must allow a user to register a new account using an email address and password.

- **REQ-AUTH-01.1:** The system must verify that the provided email address is not already linked to an existing account.
- **REQ-AUTH-01.2:** The system must display the error message "Email already in use" if a duplicate email registration is attempted.
- **REQ-AUTH-01.3:** The system must enforce a minimum password length of 8 characters.
- **REQ-AUTH-01.4:** The system must enforce the inclusion of at least one special character (e.g., @, #, $, %, ^, &, *) in the password.
- **REQ-AUTH-01.5:** The system must verify the ownership of the email address with a one-time 6-digit code sent to the registered email.
- **REQ-AUTH-01.6:** The system must expire the one-time code 10 minutes after it is issued.
- **REQ-AUTH-01.7:** The system must activate the account only after successful code verification.

REQ-AUTH-02: The system must allow a user to log in using their registered email and password.

- **REQ-AUTH-02.1:** The system must deny access and display the error message "Invalid email or password" if the provided credentials do not match database records.
- **REQ-AUTH-02.2:** The system must lock the user account for 15 minutes after 5 consecutive failed login attempts.

REQ-AUTH-03: The system must provide a "Forgot Password" function to recover lost account access.

- **REQ-AUTH-03.1:** The system must send a password reset link to the user's registered email address upon request.
- **REQ-AUTH-03.2:** The system must invalidate the password reset link 15 minutes after it is generated.
- **REQ-AUTH-03.3:** The system must enforce the standard password strength requirements (**REQ-AUTH-01.3**, **REQ-AUTH-01.4**) when the user sets a new password.

REQ-AUTH-04: The system must allow an authenticated user to securely log out.

- **REQ-AUTH-04.1:** The system must terminate the active session token upon logout and redirect the user to the login screen.

REQ-AUTH-05: The system must allow users to edit and manage their profile.

- **REQ-AUTH-05.1:** The system must allow users to update their displayed Full Name, validating that the input is non-empty and between 2 and 50 characters.
- **REQ-AUTH-05.2:** The system must allow users to delete their account
- **REQ-AUTH-05.3:** The system must prompt the user for confirmation before account deletion.
- **REQ-AUTH-05.4:** The system must permanently delete the user's account and associated data from the database once confirmed.

4.2 Preference Management

4.2.1 Description and Priority

- **High priority**. Manages the user's physical mobility constraints and environmental preferences that dynamically dictate the routing engine's weightage formula.

4.2.2 Stimulus/Response Sequences

- User navigates to profile -> User modifies mobility or route preferences -> System validates inputs -> System saves preferences to database -> System applies new weights to all subsequent route calculations.

4.2.3 Functional Requirements

REQ-PREF-01: The system must display a "Personal Declaration Form" within the user profile interface.

- **REQ-PREF-01.1:** The system must automatically pre-fill the form with the user's most recently saved database preferences upon loading.

REQ-PREF-02: The system must provide a dropdown menu for “mobility_needs” to dictate physical routing constraints.

- **REQ-PREF-02.1:** The “mobility_needs” dropdown must strictly contain the following options: None, Wheelchair, Mobility Aid.
- **REQ-PREF-02.2:** The system must default the “mobility_needs” field to "None" for newly registered accounts.

REQ-PREF-03: The system must allow the user to save their modified preferences.

- **REQ-PREF-03.1:** The system must commit the updated form data to the database under the active user's unique ID.
- **REQ-PREF-03.2:** The system must immediately apply the updated variables to the routing engine for the very next search query.

REQ-PREF-04: The system must provide a numeric input field for "Age" within the user profile interface.

- **REQ-PREF-04.1:** The system must only accept positive integers for the age field.

REQ-PREF-05: The system must provide a selection for "Heat / UV sensitivity" to dictate routing preferences regarding shaded or indoor paths.

- **REQ-PREF-05.1:** The "Heat / UV sensitivity" selection must strictly contain the following options: Low, Medium, High.
- **REQ-PREF-05.2:** The system must default this field to "Medium" for newly registered accounts.

REQ-PREF-06: The system must provide a slider input for "Max stairs per route" accepting values from 0 to 50 steps in increments of 5, defaulting to 10 steps for standard users and locked to 0 steps if “mobility_needs” is "Wheelchair".

REQ-PREF-07: The system must provide a dropdown menu for "Preferred pace" with options: "Slow" (3.0 km/h), "Relaxed" (4.0 km/h), and "Brisk" (5.5 km/h), defaulting to "Relaxed".

4.3 Search & Map Interface

4.3.1 Description and Priority

- **High priority**. Handles the primary user interface where the map is rendered, and origin/destination points are defined.

4.3.2 Stimulus/Response Sequences

- User grants GPS permission → System renders map and centers on current location → User inputs destination → System validates the coordinates.

4.3.3 Functional Requirements

REQ-MAP-01: The system must render an interactive map interface using the selected external map provider API.

- **REQ-MAP-01.1:** The system must allow the user to interactively zoom and pan the map interface.

REQ-MAP-02: The system must prompt the user for device location (GPS) permissions upon first entering the main UI.

- **REQ-MAP-02.1:** The system must auto-fill the "Origin" input field with the user's current GPS location if permission is granted.
- **REQ-MAP-02.2:** The system must leave the "Origin" input field blank for manual entry if GPS permission is denied.

REQ-MAP-03: The system must provide autocomplete suggestions for address searches using the external Search API.

REQ-MAP-04: The system must validate all input Origin and Destination coordinates against the Singapore geographical bounding box.

- **REQ-MAP-04.1:** The system must reject the routing request if the Origin coordinate falls outside the predefined Singapore bounding box.
- **REQ-MAP-04.2:** The system must display the text "Invalid location. Please select a valid origin within Singapore" if the Origin coordinate falls outside the predefined Singapore bounding box.
- **REQ-MAP-04.3:** The system must reject the routing request if the Destination coordinate falls outside the predefined Singapore bounding box.
- **REQ-MAP-04.4:** The system must display the text "Invalid location. Please select a valid destination within Singapore" if the Destination coordinate falls outside the predefined Singapore bounding box.

REQ-MAP-05: The system must display markers for the origin and destination on the map interface once selected.

4.4 Routing & Scoring Engine

4.4.1 Description and Priority

- **High priority**. The core algorithmic backend that generates multiple paths, applies the user's preferences, and ranks the results based on segment confidence.

4.4.2 Stimulus/Response Sequences

- System receives valid Origin/Destination → Engine calculates 5 routes → Engine applies profile weights → System displays sorted routes.

4.4.3 Functional Requirements

REQ-RTE-01: The routing engine must attempt to generate up to 5 distinct route variants for every valid search query.

- **REQ-RTE-01.1:** If the user's “mobility_needs” is set to "Wheelchair", the engine must strictly exclude all segments containing steps, escalators, or vertical curbs from all route variants before applying preference optimizations.
- **REQ-RTE-01.2:** The engine must generate a "Direct Route" optimized solely for the shortest physical distance.

- **REQ-RTE-01.3**: The engine must generate a "Maximum Shelter Route" maximizing covered linkways and indoor paths.
- **REQ-RTE-01.4:** The engine must generate a "Barrier-Free Route" containing zero stairs, escalators, or curbs.
- **REQ-RTE-01.5:** The engine must generate a "Low-Effort Route" minimizing steep elevation changes.
- **REQ-RTE-01.6:** The engine must generate a "Wide/Quiet Route" maximizing Park Connector Network (PCN) usage.
- **REQ-RTE-01.7:** The system must return a “No route available” payload for a specific route variant if no path satisfies that variant's unique physical constraints, while continuing to generate the remaining viable variants.

REQ-RTE-02: The system must process routing metadata for each generated route.

- **REQ-RTE-02.1:** The routing engine must calculate the total estimated walking time.
- **REQ-RTE-02.2:** The routing engine must calculate the total physical distance.
- **REQ-RTE-02.3:** The routing engine must calculate the sheltered distance percentage.
- **REQ-RTE-02.4:** The system must append the calculated walking time to the route response payload.
- **REQ-RTE-02.5:** The system must append the calculated distance to the route response payload.
- **REQ-RTE-02.6:** The system must append the calculated sheltered percentage to the route response payload.
- **REQ-RTE-02.7:** The system must display "Distance information not available" or "Estimated travel time not available" if the respective data is incomplete.

REQ-RTE-03: The routing engine must filter out any map segment possessing a confidence score less than -10 when generating the "Barrier-Free Route".

REQ-RTE-04: The system must calculate a final match score on a scale of 0 to 100 for each route using the user's saved mobility profile (“mobility_needs”, “age”, “heat_uv_sensitivity”, “max_stairs_per_route”, and “preferred_pace”).

REQ-RTE-05: The system must present the 5 routes to the user in a UI drawer.

- **REQ-RTE-05.1:** The system must sort the displayed routes in descending order based on their match score.
  - **REQ-RTE-05.1.1:** The system must resolve identical match scores by prioritizing the route with the shorter estimated walking time.
  - **REQ-RTE-05.1.2:** The system must prioritize the route with the higher sheltered distance percentage if the estimated walking times are also tied.
- **REQ-RTE-05.2:** The system must visually highlight the route with the highest match score with a "Starred" badge.

4.5 Live Navigation

4.5.1 Description and Priority

- **High priority**. Provides the active turn-by-turn guidance once a user selects a route.

4.5.2 Stimulus/Response Sequences

- User selects a route → System initiates tracking → User moves → System updates instructions → User reaches destination → System updates segment confidence.

4.5.3 Functional Requirements

REQ-NAV-01: The system must anchor the map view to the user's real-time GPS coordinate during active navigation.

REQ-NAV-02: The system must display a text-based directional instruction.

- **REQ-NAV-02.1:** The system must dynamically update the text instruction based on the user's progression along the route polyline.

REQ-NAV-03: The system must monitor the user's deviation from the active route.

- **REQ-NAV-03.1:** The system must trigger an automatic route recalculation for current active route type if the user's GPS coordinate deviates from the active path by more than 20 meters for 5 consecutive seconds.
- **REQ-NAV-03.2:** The system must take the current location of the user as the new origin during recalculation.

REQ-NAV-04: The system must detect when the user arrives at the destination.

- **REQ-NAV-04.1:** The system must consider the route completed when the user's GPS coordinate is within a 5-meter radius of the destination point.
- **REQ-NAV-04.2:** The system must automatically increment the confidence score of all traversed map segments by +1 upon successful route completion, up to a maximum clamp limit of +20.

REQ-NAV-05: The system must automatically update the metric information (remaining distance and estimated time) as the user travels through the route.

4.6 Community Hazard & Local Feed

4.6.1 Description and Priority

- **Medium priority**. Handles the crowdsourced data loop, allowing users to report unwalkable checkpoints and view local warnings.

4.6.2 Stimulus/Response Sequences

- User taps “Live Report” → System logs hazard → System updates database → System displays hazard on the local feed for nearby users.

4.6.3 Functional Requirements

REQ-HAZ-01: The system must display a "Live Report" button on the screen during active navigation.

REQ-HAZ-02: The system must allow the user to report a map segment as "Unwalkable", “Path blocked”, “Flooding”, “Something Else”, “Lift down” and “Broken Surface” via the Live Report button.

- **REQ-HAZ-02.1:** The system must prompt the user for confirmation before submitting the report.
- **REQ-HAZ-02.2:** The system must map the hazard to the specific route segment closest to the user's current GPS coordinate.
- **REQ-HAZ-02.3:** The system must deduct 2 points from that specific map segment's confidence score in the database upon report confirmation, down to a minimum clamp limit of -20.

REQ-HAZ-03: The system must provide a dedicated "News/Hazards" screen.

- **REQ-HAZ-03.1:** The system must query the database for all map segments have received a confirmed hazard report within the past 72 hours, or with a negative confidence score.
- **REQ-HAZ-03.2:** The system must filter the queried hazards to only include those located within a strictly 5-kilometer radius of the user's current GPS location.
- **REQ-HAZ-03.3:** The system must display the filtered list of hazards to the user.
- **REQ-HAZ-03.4:** The system must sort the displayed hazard list chronologically, showing the most recently reported hazards at the top.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system must render the 5 calculated route variants on the map interface within 3.0 seconds of the user submitting a valid destination over a standard 4G connection.
- The system must display address autocomplete suggestions within 500 milliseconds of user keystroke input during search.
- The system must update the user's live GPS location marker on the map with a maximum latency of 1.0 second from receiving the device's signal.
- The live hazard feed must successfully load within 3.0 seconds of the user navigating to the News screen.

5.2 Safety Requirements

- The routing engine shall expire and delete submitted hazardous obstructions and live deductions 72 hours after submission unless confirmed by a subsequent report.
- The system must suppress interactive map interactions and display a motion warning prompt if the device moves faster than 25 km/h during active navigation.
- The system must fail safely with a 100% rejection rate for any route calculation originating or terminating outside Singapore's coordinate boundary, completely preventing unmapped routing in unsupported international border areas.

5.3 Security Requirements

- The backend database must encrypt all stored user passwords using a cryptography hashing algorithm.
- The backend must automatically invalidate a user's active session token after 30 days of inactivity.
- The backend API must sanitize all text inputs against SQL injection and cross-site scripting (XSS) before query execution.

5.4 Software Quality Attributes

- **TESTABILITY:** The routing engine must return an identical route payload when queried with identical Origin, Destination, and preference variables over 100 consecutive requests.
- **ROBUSTNESS:** The app must reload the active navigation screen in 1.5 seconds or less when the user switches back to it from another app.
- **ADAPTABILITY:** The system must apply changes to the confidence score configuration file without requiring a source code recompilation.
- **MAINTAINABILITY:** The routing service must isolate external map provider integrations behind an API adapter interface so changing base map endpoints requires zero alterations to core scoring logic.
- **USABILITY:** At least 95% of first-time users must successfully complete initial profile setup and initiate their first route navigation on their first attempt without external assistance.

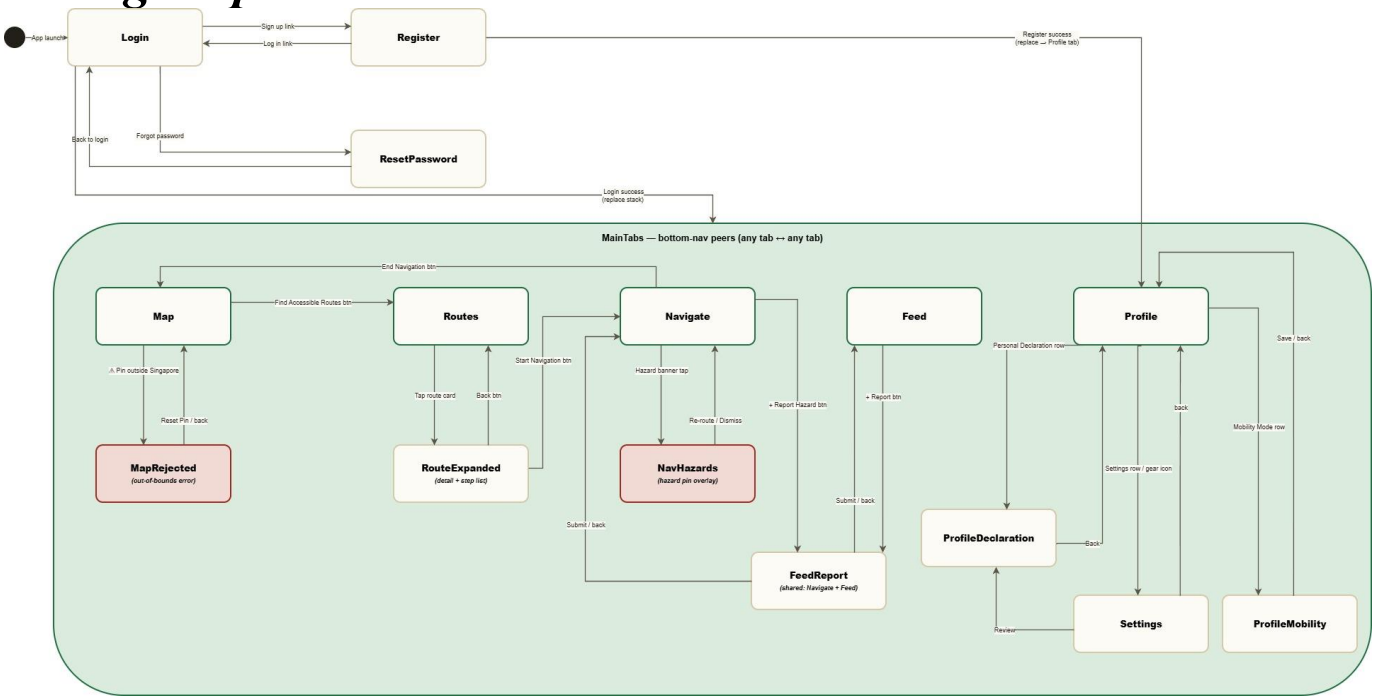
5.5 Business Rules

- The system must restore a map segment's confidence score to its default baseline exactly 72 hours after a single "Live Report" deduction.
- The system must grant profile editing privileges exclusively to the user account matching the profile's unique User ID.
- The system must define weights and 0–100 component scores where:
    - S_{eff} is distance efficiency = min(100, 100 × ($Distance_{Direct}$ / $Distance_{Route}$));
    - $S_{shelter}$ is sheltered distance percentage with weight $w_{shelter}$ (0.20 for High UV, 0.15 for Medium, 0.10 for Low);
    - S_{effort} is 100 − 2 × stairs_count with weight w_{effort} (0.20 if age ≥ 65 or mobility aid used, else 0.10);
    - S_{conf} is ((avg_confidence + 20) / 40) × 100 with weight w_{conf} = 0.15; and w_{eff} = 0.25.
    - All weights are normalised so their sum equals 1.0.
- The system must compute each route's match score (0 to 100) using the formula: Match_Score = G × (w_{eff} × S_{eff} + $w_{shelter}$ × $S_{shelter}$ + w_{effort} × S_{effort} + w_{conf} × S_{conf}), where gate G is 0 if stairs or curbs violate wheelchair constraints, and 1 otherwise.

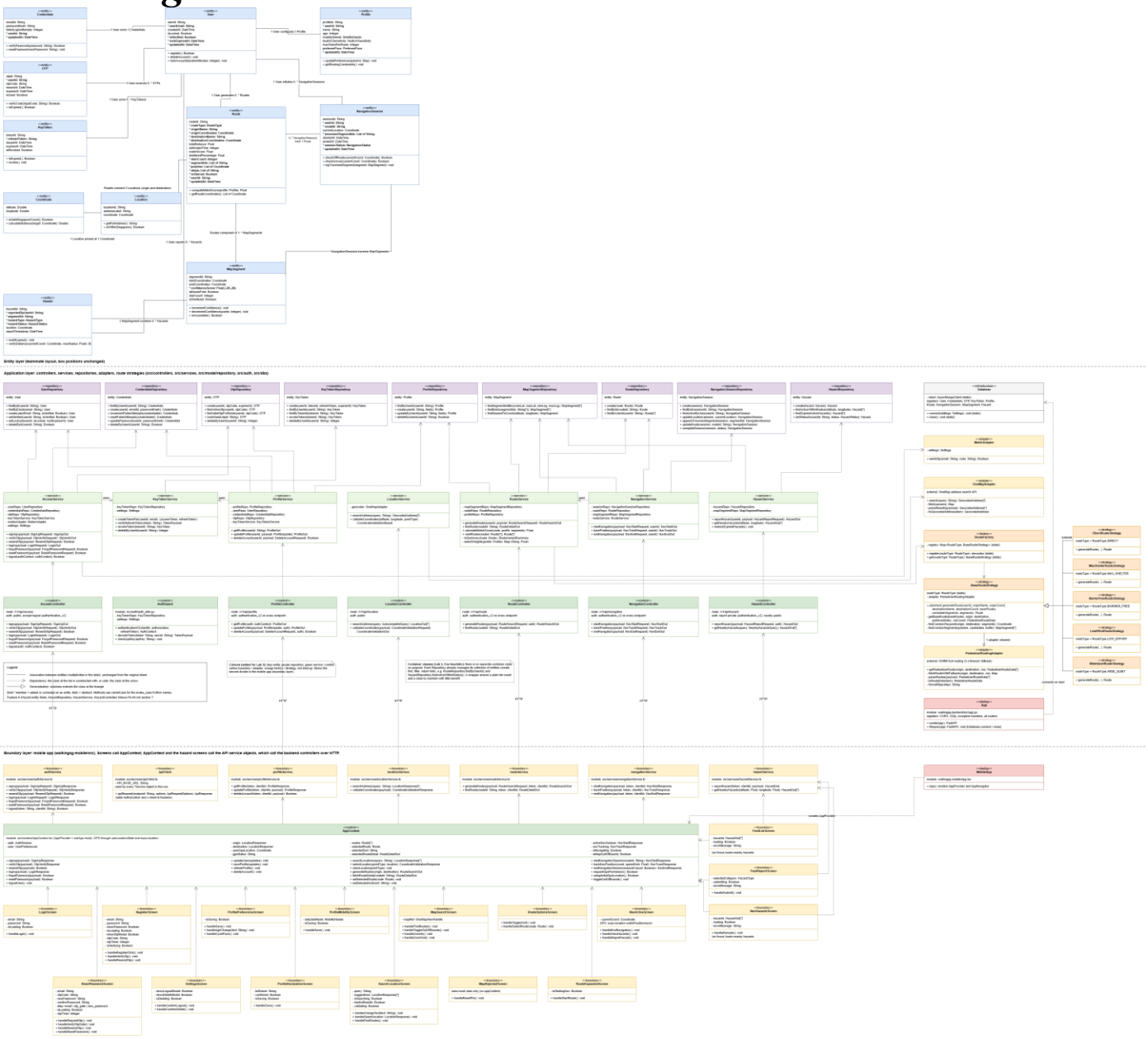
6. Other Requirements

Appendix A: Analysis Models

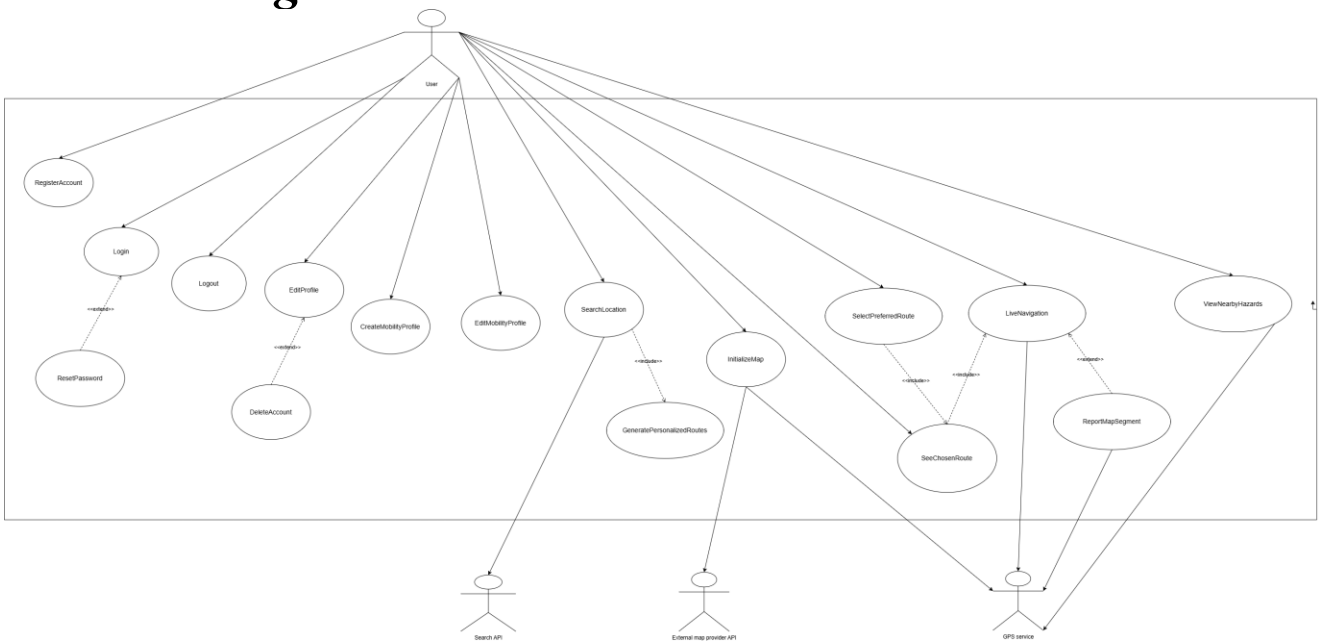
Dialog map



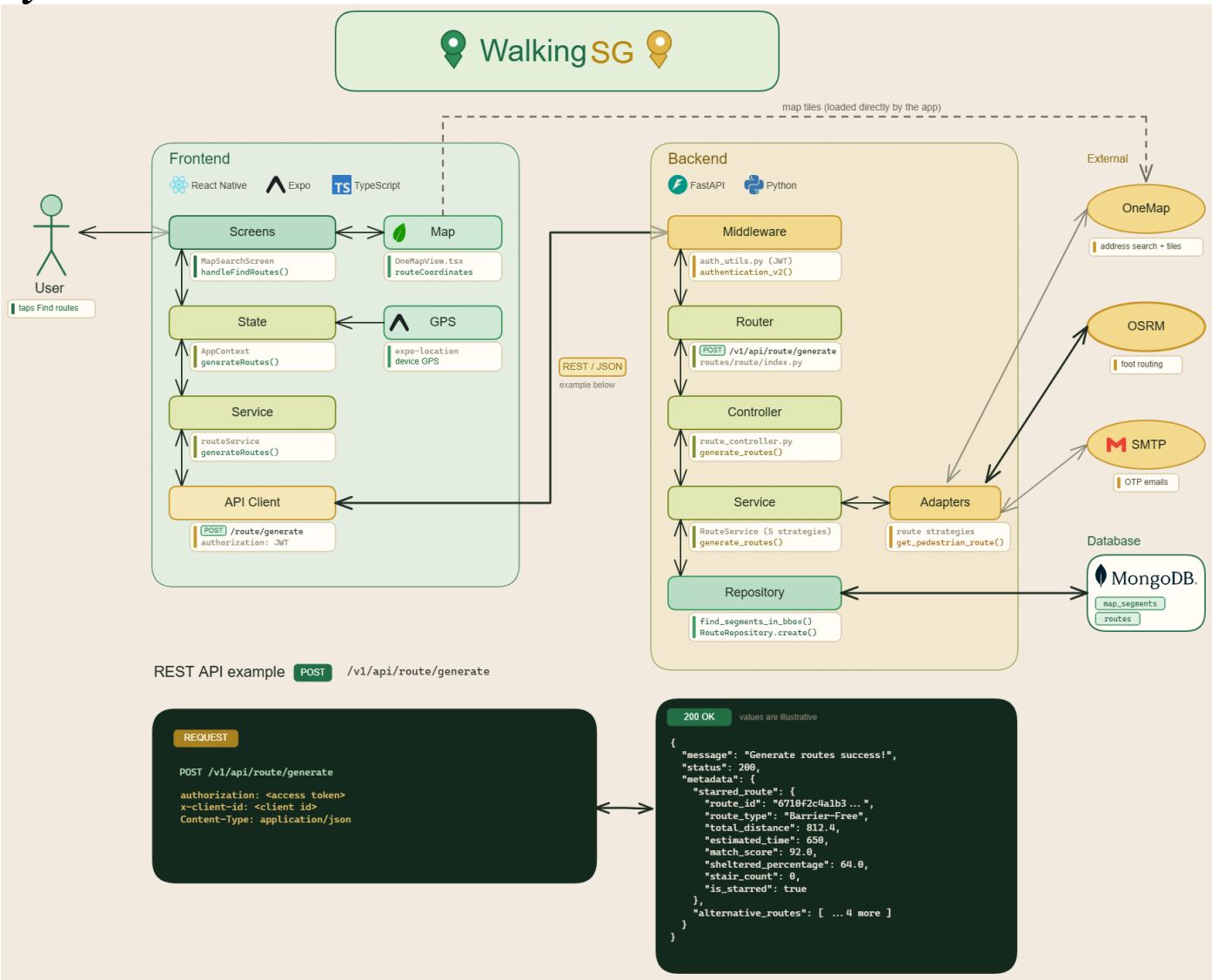
Class diagram



Use case diagram



System Architecture



Appendix B: To Be Determined List

<Collect a numbered list of the TBD (to be determined) references that remain in the SRS so they can be tracked to closure.>

Appendix C: Data Dictionary

Term	Definition	Attribute	Relationships
User	**A registered person with an account and credentials logged into the app's database. The user will also have a profile which will have all their preferences according to their needs.**	**user_id, email, createdAt, isLocked**	One-to-One: **1 User has exactly 1 set of Credentials** One-to-One: **1 User has exactly 1 Profile** One-to-Many: **1 User can generate multiple Routes** One-to-Many: **1 User can report multiple Hazards** One-to-Many: **1 User can hold multiple Session Tokens** One-to-Many: **1 User can be issued multiple OTPs across resend requests.**
Credentials	**A combination of email and**	**user_id, email_id,**	One-to-One: **1 set of**

Term	Definition	Attribute	Relationships
	password to authenticate access into the user's account. Credentials are created on registration for the app. Credentials can be reset securely with a link if the user forgets them to regain access.	**passwordHash, failedLoginAttempts**	**Credentials belongs to exactly 1 User**
Navigation Session	**A session is created when the user confirms a route via the "Proceed" action and remains active until the user arrives at the destination or manually cancels navigation. It tracks the user's real-time position against the active route to detect arrival, deviation, and segment traversal for confidence score updates.**	**Session_id, user_id, route_id, started_at, ended_at, status, segments_traversed**	Many-to-One: **Many Navigation Sessions can belong to 1 User over time** One-to-One: **1 active Navigation Session references exactly 1 Route.** Many-to-Many: **1 Navigation Session traverses many Map Segments** Many-to-One: **Many Navigation Sessions can be authorized under 1 Session Token across its active lifetime.**
Session Token	**A cryptographically generated string issued to a user upon successful login. It represents and authorises an active authenticated session. A token is invalidated on explicit logout, password reset, or after 30 days of inactivity.**	**token_id, user_id, issued_at, expires_at, last_active_at, is_revoked**	One-to-Many: **1 Session Token can authorize multiple Navigation Sessions over its lifetime.** Many-to-One: **Many Session Tokens can belong to 1 User**
OTP	**A system-generated 6-digit numeric code issued to a user's registered email address to verify email ownership during account registration. It is strictly single-use and automatically expires 10 minutes after issuance.**	**otp_id, user_id, otp_code, issued_at, expires_at, is_used**	Many-to-One: **Many OTPs can be issued to 1 user across multiple resend requests**
Profile	**Users' data that they have**	**profileId, name,**	One-to-One: **1 Profile**

Term	Definition	Attribute	Relationships
	provided to personalise their account. Data includes preferences, personal details and past usage data. Data can be updated by profile when necessary.	**mobility_needs, age, heat_uv_sensitivity, name, max_stairs_per_route, preferred_pace, max_duration**	**belongs to exactly 1 User** One-to-Many: **1 Profile weights the ranking of multiple Routes**
Location	**Users' location which is obtained via device's GPS location. This can only be obtained when the user provides permission to access this data.**	**location_id, coordinate, address_label**	One-to-One: **1 User has at most 1 Location at a time** One-to-One: **1 Location can be used as the Origin of 1 Route** One-to-One: **1 Location is pinned at 1 Coordinate** Two-to-Many: **2 Locations can be connected by many Routes**
Coordinate	**A set of numbers or letters that pinpoint an exact location on a map, GPS.**	**latitude, longitude**	One-to-One: **1 Coordinate represents 1 Location**
Origin	**The user's starting location which is input to compute a route. Location can be obtained either via autofilling current location or obtaining manual input from the user directly.**	**coordinate, address_label**	One-to-One: **1 Route starts from exactly 1 Origin**
Destination	**The end location for the trip where the user wishes to reach. The location can be obtained via autofilling current location or receiving manual input from the user directly.**	**destination_id, coordinate, address_label**	One-to-Many: **1 User can search for multiple Destinations.** One-to-One: **1 Route computes towards exactly 1 Destination.**

Term	Definition	Attribute	Relationships
Route	**The computer path from origin to destination which is generated based on the routing algorithm taking into account the total length of the journey and the user's preferences.**	**user_id, route_id, route_type, total_distance, estimated_time, match_score, sheltered_percentage**	One-to-Many: **1 User can generate multiple Routes.** Many-to-Two: **Many Routes can connect 2 locations** Many-to-Many: **One Routes can be composed of many Map segments.**
Map segments	**An individual stretch of walkable path that routes are built from and against which confidence is recorded.**	**segment_id, start_coordinates, end_coordinates, confidence_score, is_barrier_free, stair_count, is_sheltered**	Many-to-Many: **One Map segment can be shared across Many computed Routes.** One-to-Many: **1 Map segment can contain multiple reported Hazards.** Many-to-Many: **1 Map Segment can be traversed across by many Navigation Sessions**
Hazard	**A hazard is an object or obstruction to a map segment, which could be a part of a computed route for a user, that can cause harm to the user or prevent them from reaching their destination via the route.**	**hazard_id, hazard_type, hazard_status, location, reported_by_user_id, report_timestamp**	One-to-Many: **1 User can report multiple Hazards.** One-to-Many: **1 Map segment can contain multiple Hazards.**

Appendix D: Use case description

6.1 Use Case Descriptions

#1.1: Authentication

Use Case ID:	1.1-1		
Use Case Name:	RegisterAccount		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Allows a new user to register an account using an email and password, verified via an email one-time code.
Preconditions:	The user has installed and launched the WalkingSG application and is on the registration interface.
Postconditions:	A new user account is successfully activated in the database, and the user is redirected to the login interface.
Priority:	High
Frequency of Use:	Low (once per user)
Flow of Events:	1. The user enters their email address, sets a password, and clicks "Register". 2. System validates that the password has greater or equal than 8 characters and includes at least one special character. 3. The system verifies that the provided email address is not already linked to an existing account in the database. 4. The system generates a 6-digit one-time code and sends it to the user's email address. 5. The user inputs the received 6-digit verification code into the verification screen and clicks "Verify". 6. The system validates the code, activates the account in the database, and navigates the user to the login screen.
Alternative Flows:	**AF-1 (Resend Verification Code):** At step 5, the user clicks "Resend Code". The system invalidates the previous code, generates a new 6-digit verification code, sends it to the user's email, and restarts the 10-minute expiry timer.

Exceptions:	**EX-1 (Weak Password):** At step 2, if the password is shorter than 8 characters or lacks a special character (@, #, $, %, ^, &, *), the system displays "Password must be at least 8 characters and contain at least one special character" and prompts the user to re-enter. **EX-2 (Duplicate Email):** At step 3, if the email already exists in

	the database, the system displays "Email already in use" and aborts registration. **EX-3 (Expired or Invalid OTP):** At step 6, if the 6-digit code is incorrect or exceeds the 10-minute expiration window, the system displays "Invalid or expired code" and prompts the user to re-enter or request a new code
Includes:	None
Special Requirements:	Passwords must be hashed using cryptography algorithm before database storage. The OTP must strictly expire 10 minutes after issuance (**REQ-AUTH-01.6**).
Assumptions:	The user has access to a functional email inbox to receive external mail.
Notes and Issues:	None

Use Case ID:	1.1-2		
Use Case Name:	Login		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Authenticates a registered user using their email and password to grant access to the application.
Preconditions:	The user possesses an activated WalkingSG account and is not currently locked out.
Postconditions:	An active session token is issued, and the user is authenticated into the application.
Priority:	High
Frequency of Use:	High (daily/weekly)
Flow of Events:	1. The user enters their registered email address and password on the login screen. 2. The user selects the "Log In" button. 3. checks if the account is currently locked due to prior failed attempts. 4. The system validates the credentials against stored database records using cryptography verification. 5. The system establishes an active session token and navigates the user to the main map screen.
Alternative Flows:	**AF-1 (Forgot Password Path):** At step 1, the user selects "Forgot Password". The system navigates the user to the password recovery screen (triggers Use Case #1.1-3).
Exceptions:	**EX-1 (Invalid Credentials):** At step 4, if the email and password do not match database records, the system increments the failed login counter, denies access, and displays "Invalid email or password". **EX-2 (Account Lockout Triggered):** At step 4, upon the 5th consecutive failed login attempt, the system locks the account for 15 minutes and displays "Account locked for 15 minutes due to multiple failed attempts". **EX-3 (Account Already Locked):** At step 3, if the user attempts login while the 15-minute lockout timer is active, the system

	denies entry and displays "Account locked. Please try again later".
Includes:	None
Special Requirements:	Failed login attempts must be tracked in server-side cache with a 15-minute expiry time. Passwords must be masked on screen by default
Assumptions:	The user device has a stable internet connection.
Notes and Issues:	None

Use Case ID:	1.1-3		
Use Case Name:	ResetPassword		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Allows a registered user who forgot their password to regain account access via a time-limited email reset link.
Preconditions:	The user has navigated to the "Forgot Password" screen from the login interface.
Postconditions:	The user's account password is updated in the database, and old credentials/tokens are invalidated.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. The user enters their registered email address and clicks "Send Reset Link". 2. System validates email existence, generates a secure reset token, and emails the link to the user 3. The user clicks the reset link in the email to open the password reset interface. 4. The user enters and confirms a new password that satisfies complexity rules (at least 8 characters and at least one special character). 5. The system validates the password, updates the database, invalidates the reset link, and redirects to the login screen.
Alternative Flows:	**AF-1 (User Aborts):** At step 1 or step 4, the user navigates back to the login screen. The system cancels the operation without updating credentials
Exceptions:	**EX-1 (Expired Reset Link):** At step 3, if the user opens the reset link more than 15 minutes after generation, the system displays "Password reset link has expired" and prompts the user to request a new link. **EX-2 (Password Fails Complexity):** At step 5, if the new password has fewer than 8 characters or lacks a special character, the system displays "Password does not meet requirements" and prompts for re-entry.
Includes:	None

Special Requirements:	Reset links must be single-use and invalidated immediately upon successful password change or after 15 minutes (**REQ-AUTH-03.2**).
Assumptions:	Mail delivery service operates within acceptable latency (< 60 seconds).
Notes and Issues:	None

Use Case ID:	1.1-4		
Use Case Name:	Logout		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Terminates the user's active session and securely logs them out of the application
Preconditions:	The user is currently authenticated and logged in.
Postconditions:	The session token is permanently invalidated, and the application returns to the unauthenticated login screen.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. User navigates to the profile/settings interface and clicks "Log Out". 2. System prompts the user with a dialogue to confirm the logout action. 3. User confirms the logout action. 4. System invalidates the active session token, clears local user cache, and redirects the user to the login screen.
Alternative Flows:	**AF-1 (Cancel Logout):** At step 3, the user selects "Cancel". The system dismisses the prompt and maintains the active session on the current screen
Exceptions:	**EX-1 (Token Revocation Error):** At step 4, if the server fails to process session revocation due to connectivity drop, the system forcefully clears local tokens from device storage and redirects to the login screen.
Includes:	None
Special Requirements:	Must terminate session tokens immediately to prevent replay or unauthorised access.
Assumptions:	None
Notes and Issues:	None

Use Case ID:	1.1-5		
Use Case Name:	EditProfile		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Allows a user to view and update their personal account information, specifically their display name.
Preconditions:	The registered user is logged in and viewing their user profile
Postconditions:	The user's updated personal information is saved to the database.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. User selects "Edit Profile" on the profile interface. 2. System displays an editable form pre-populated with the user's current registered name and email. 3. User updates their name field and clicks "Save Changes". 4. System validates that the name input is non-empty and adheres to format constraints of length between 2 and 50 characters. 5. System commits the updated name to the database and displays "Profile updated successfully".
Alternative Flows:	**AF-1 (Discard Edits):** At step 3, the user clicks "Cancel" or taps the back icon. The system discards all modifications and restores the original profile view without updating the database.
Exceptions:	**EX-1 (Blank Name Field):** At step 4, if the user clears the name field and attempts to save, the system displays "Name cannot be left blank" and retains focus on the input field.
Includes:	None
Special Requirements:	In accordance with Business Rule 2.4, editing privileges must be restricted strictly to the user matching the unique User ID.
Assumptions:	Registered email address cannot be edited via the standard name edit form (managed separately under authentication).
Notes and Issues:	None

Use Case ID:	1.1-6		
Use Case Name:	DeleteAccount		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Allows a user to permanently delete their account and remove all personal records from the system.
Preconditions:	The user is authenticated and navigating the account settings interface.
Postconditions:	The user account, credentials, and associated personal profile data are permanently erased from the database.
Priority:	Medium
Frequency of Use:	Very Low
Flow of Events:	1. The user selects "Delete Account" on the settings/profile screen. 2. The system displays a confirmation dialog warning that deletion is permanent and irreversible. 3. The user confirms the deletion request. 4. The system permanently deletes the user's account records, preferences, and personal information from the database. 5. The system terminates all active sessions and redirects the user to the welcome screen.
Alternative Flows:	**AF-1 (Cancel Deletion):** At step 3, the user selects "Cancel" on the confirmation prompt. The system dismisses the prompt and leaves the user account untouched
Exceptions:	**EX-1 (Database Deletion Failure):** At step 4, if a database timeout occurs, the system displays "Unable to delete account at this time. Please try again later" and retains all user data.
Includes:	None
Special Requirements:	Full data purge must comply with Singapore PDPA right-to-erasure and **REQ-AUTH-05.4**.
Assumptions:	None
Notes and Issues:	None

#1.2: Preference Management

Use Case ID:	1.2-1		
Use Case Name:	CreateMobilityProfile		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Allows a newly registered user to set up their initial mobility constraints and environmental routing preferences for the first time.
Preconditions:	The user has registered and logged in and has not yet customised their default mobility profile.
Postconditions:	An initial personalised preference profile is established in the database under the user ID and bound to the routing engine.
Priority:	High
Frequency of Use:	Low (strictly once per user upon onboarding)
Flow of Events:	1. User navigates to the "Personal Declaration Form" in the profile section for the first time. 2. System initialises the form with default baseline values (mobility_needs as "None", Heat / UV sensitivity as "Medium"). 3. User configures their “age”, initial “mobility_needs” constraint, “heat_uv_sensitivity”, “max_stairs_per_route”, and “preferred_pace”. 4. User selects "Save Preferences" to confirm initial setup. 5. System commits the initial profile record to the database, activates routing engine binding, and displays "Profile created successfully".
Alternative Flows:	**AF-1 (Retain System Defaults):** At step 3, the user chooses to keep all default settings ("None" mobility_needs, "Medium" UV sensitivity) and selects "Save Preferences". The system records the default profile without modifications.
Exceptions:	**EX-1 (Setup Interruption):** At step 3, if the application loses connection before saving, the system preserves default parameters in the local session and prompts the user to submit again upon reconnection.
Includes:	None

Special Requirements:	Must comply with Singapore PDPA regulations regarding the collection and storage of user mobility and physical handicap data.
Assumptions:	Newly created user accounts are automatically initialised with database default values per **REQ-PREF-02.2** and **REQ-PREF-05.2**.
Notes and Issues:	None

Use Case ID:	1.2-2		
Use Case Name:	EditMobilityProfile		
Created By:	Lin Ziqi	Last Updated By:	Lin Ziqi
Date Created:	03/09/2026	Date Last Updated:	03/09/2026

Actor:	User
Description:	Allows an existing user to update their previously saved mobility constraints and routing preferences.
Preconditions:	The user has previously registered and configured an active mobility profile (Use Case #1.2-1 completed).
Postconditions:	Updated preferences are committed to the database, cached route data is flushed, and the routing engine immediately applies new parameters.
Priority:	Medium
Frequency of Use:	Low to Medium (occasional updates when physical condition or mobility constraint changes)
Flow of Events:	1. User opens the "Personal Declaration Form" in the profile tab. 2. System automatically pre-fills all form controls with the user's most recently saved database preferences. 3. User modifies their “mobility_needs”, “age”, “heat_uv_sensitivity”, “max_stairs_per_route”, or “preferred_pace”. 4. User selects "Save Preferences". 5. System commits the modified data to the database under the user ID. 6. System flushes locally cached route preferences, updates the routing engine parameters, and displays "Preferences updated successfully".
Alternative Flows:	**AF-1 (Discard Modifications):** At step 3, the user leaves the screen without saving. The system discards unconfirmed changes and keeps existing preferences intact.
Exceptions:	**EX-1 (Database Write Failure):** At step 5, if the database transaction fails, the system displays "Failed to update preferences. Please try again", retaining existing parameters.
Includes:	None

Special Requirements:	**REQ-PREF-05.2:** The system must immediately clear any locally cached route preferences and apply the updated variables to the routing engine for the very next query.
Assumptions:	Changes take effect immediately without requiring the user to log out and log in again.
Notes and Issues:	None

#1.3 Search & Map Interface

Use Case ID:	1.3-1		
Use Case Name:	InitializeMap		
Created By:	Benjamin	Last Updated By:	Benjamin
Date Created:	9/4/2026	Date Last Updated:	9/4/2026

Actor:	User
Description:	Renders the interactive map interface and handles initial GPS location permission prompts to auto-fill or clear the “Origin” input.
Preconditions:	Application is opened; Map API service is available.
Postconditions:	Interactive map is rendered, and initial “Origin” input field state is set.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System renders interactive map interface using external map provider API. 2. System allows user to zoom and pan on map interface. 3. System prompts user for device location (GPS) permissions upon entering main UI.
Alternative Flows:	**AF-1 (GPS Permission Granted):**System auto-fills "Origin" input field with user's current GPS location. **AF-2 (GPS Permission Denied):** System leaves "Origin" input field blank for manual entry.
Exceptions:	None
Includes:	None
Special Requirements:	Interactive map zooming and panning support (**REQ-MAP-01.1**).
Assumptions:	Device hardware supports GPS capabilities.
Notes and Issues:	None

Use Case ID:	1.3-2		
Use Case Name:	SearchLocation		
Created By:	Benjamin	Last Updated By:	Benjamin
Date Created:	9/4/2026	Date Last Updated:	9/4/2026

Actor:	User
Description:	Provides autocomplete suggestions for address searches and validates input coordinates against the Singapore bounding box before displaying markers.
Preconditions:	Map interface is loaded.
Postconditions:	Valid origin and destination coordinates are validated and displayed as markers on the map.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user inputs a search query into Origin/Destination fields. 2. The system provides autocomplete suggestions using external Search API. 3. The user selects origin and destination addresses. 4. System validates Origin and Destination coordinates against Singapore geographical bounding box. 5. The system displays markers for origin and destination on map interface. 6. The system calls **GeneratePersonalizedRoutes** to generate route variants based on the validated Origin and Destination.
Alternative Flows:	None
Exceptions:	**EX-1(Origin Coordinate Outside Bounding Box):** System rejects routing request and displays "Invalid location. Please select a valid origin within Singapore". **EX-2(Destination Coordinate Outside Bounding Box):** System rejects routing request and displays "Invalid location. Please select a valid destination within Singapore".
Includes:	**GeneratePersonalizedRoutes**
Special Requirements:	Hard geographical boundary checking restricted strictly to Singapore bounding box.
Assumptions:	External Search API is reachable.
Notes and Issues:	None

#1.4 Routing & Scoring Engine:

Use Case ID:	1.4-1		
Use Case Name:	GeneratePersonalizedRoutes		
Created By:	Huy	Last Updated By:	Huy
Date Created:	04/09/2026	Date Last Updated:	05/09/2026

Actor:	User (Internal Use Case of SearchLocation)
Description:	The system retrieves the user's routing preferences, processes validated coordinates to generate exactly 5 distinct route variants, calculates their metadata, and scores them.
Preconditions:	The user has submitted a valid Origin and Destination coordinate.
Postconditions:	A complete payload containing 5 scored route variants is returned to the UI.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The system receives the validated Origin and Destination coordinates. 2. The system retrieves the user's saved “heat_uv_sensitivity” and “mobility_needs” from the database, applying the defaults ("Medium" and "None") if no modifications exist. 3. The system generates exactly 5 distinct route variants: “Direct”, “Maximum Shelter”, “Barrier-Free”, “Low-Effort”, and “Wide/Quiet”. 4. The system calculates the total estimated walking time, and total physical distance for each route variant. 5. The system calculates a final match score for each route using the preferences retrieved in Step 2. 6. The system appends all calculated metadata and match scores to the route response payload and returns it to the client.
Alternative Flows:	**AF-1 (Hazard Filter):** At step 3, if the engine detects a map segment with a confidence score below -10, it filters that segment out of the "Barrier-Free Route" generation.
Exceptions:	**EX-2 (No Route):** At step 3, if no physical path exists between the origin and destination, the system returns a "No route available" error payload.
Includes:	None

Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

Use Case ID:	1.4.2		
Use Case Name:	SelectPreferredRoute		
Created By:	Fujita Natsuo	Last Updated By:	Fujita Natsuo
Date Created:	02/09/2026	Date Last Updated:	04/09/2026

Actor:	User
Description:	The user reviews the ranked route options and selects one for navigation.
Preconditions:	The 5 route variants carry final match scores
Postconditions:	A single route is locked in as the active navigation path.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The system displays the 5 route variants in the UI drawer, sorted by descending score 2. The system visually highlights the highest scoring route with a “Starred” badge 3. The user reviews the route and selects their preferred route.
Alternative Flows	**AF-1 (Tie-Breaker Sorting):** At step 1, if two or more routes share the same match score, the system prioritizes the route with the shorter estimated walking time. If still tied, it prioritizes the higher sheltered distance percentage. If both remain equal, the system defaults to the priority order: Maximum Shelter, Barrier-Free, Low-Effort, Wide/Quiet **AF-2 (Incomplete Metadata):** At step 1, if a route is missing distance or time data, the system displays "Distance information not available" or "Estimated travel time not available" on that specific card.
Exceptions	None
Includes	None
Special Requirements	None
Assumptions	The UI drawer can render all 5 route cards without requiring additional pagination.

Notes and Issues	None

Use Case ID:	1.4.3		
Use Case Name:	SeeChosenRoute		
Created By:	Huy	Last Updated By:	Huy
Date Created:	11/09/2026	Date Last Updated:	11/09/2026

Actor:	User
Description:	Displays the selected route as a full-screen polyline overlay on the map, allowing the user to visually inspect the path and direction before committing to live navigation.
Preconditions:	- The user has selected one route variant from the route results list (**Use Case #1.4-2** completed) - A valid route polyline payload exists for the selected route
Postconditions:	The selected route polyline is rendered on the map and the user either proceeds to live navigation or returns to the route list
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user taps on a route card from the route results list 2. The system renders the selected route as a polyline overlay on the full-screen map view, with origin and destination markers displayed 3. The system displays a summary panel showing the route's estimated time, total distance, sheltered percentage, stair count, and match score 4. The user taps the "Proceed" button to confirm the route selection 5. The system initiates live navigation for the confirmed route (**Use Case #1.5-1**)
Alternative Flows	**AF-1 (Return to Route List):** At step 4, the user taps the back icon. The system dismisses the route preview and returns the user to the route results list with all 5 route cards intact
Exceptions	**EX-1 (Route Payload Missing):** At step 2, if the route polyline data is unavailable or corrupted, the system displays "Route preview unavailable. Please try again" and

	returns the user to the route results list.
Includes	**LiveNavigation (#1.5-1)** – triggered upon user confirmation via the “Proceed” button
Special Requirements	The route polyline must be rendered within 1.0 second of the user tapping the route card
Assumptions	The route polyline coordinates are included in the route payload returned by **GeneratePersonalizedRoutes (#1.4-1)**
Notes and Issues	None

#1.5 Live Navigation

Use Case ID:	1.5-1		
Use Case Name:	LiveNavigation		
Created By:	Benjamin	Last Updated By:	Benjamin
Date Created:	04/09/2026	Date Last Updated:	04/09/2026

Actor:	User
Description:	Tracks user's real-time position during active navigation, provides dynamic directional text updates and trip metrics, automatically recalculates routes upon off-path deviation, and logs segment completion confidence scores upon destination arrival.
Preconditions:	Active route generated, GPS tracking enabled on device.
Postconditions:	Route marked as completed upon arrival within 5 meters, and confidence scores incremented.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System anchors map view to user's real-time GPS coordinate during active navigation. 2. The system displays text-based directional instruction and dynamically updates it based on user progression along route polyline. 3. The system continually monitors the user's deviation relative to the active route. 4. The system dynamically updates metric information (remaining distance and estimated time) as the user travels. 5. The system detects arrival when the user's GPS coordinate is within a 5-meter radius of the destination point. 6. System considers route completed and automatically increments confidence score of all traversed map segments by +1.
Alternative Flows:	None
Exceptions:	**EX-1(User Deviates > 20 Meters for 5 consecutive seconds):** System triggers automatic route recalculation using current GPS position as the new origin point.

Includes:	None
Special Requirements:	Dynamic update loop for distance/time metrics. Off-route threshold = 20 meters. Arrival radius = 5 meters. Increment value = +1.
Assumptions:	Device maintains active GPS tracking connection throughout navigation.
Notes and Issues:	None

#1.6 Community Hazard and Local Feed

Use Case ID:	1.6-1		
Use Case Name:	Report Map Segment		
Created By:	Hemanth	Last Updated By:	Hemanth
Date Created:	04/09/2026	Date Last Updated:	04/09/2026

Actor:	User
Description:	Allows a user to report a map segment as “Unwalkable” via the “Live Report” button.
Preconditions:	The system must have access to the user’s current GPS coordinate to identify the specific map segment that may be affected.
Postconditions:	The system must log the hazard directly to the “News/Hazards” screen. The affected map segment must have its confidence score decremented by 2 points.
Priority:	High
Frequency of Use:	Low - Medium (occurrence of hazards should not be too often)
Flow of Events:	1. The user clicks the “Live Report” button on his screen during active navigation. 2. The user reports his current location as “Unwalkable”, “Path blocked”, “Flooding”, “Something Else”, “Lift down” or “Broken Surface” on the app. 3. The system prompts the user for confirmation before submitting the report 4. The user confirms and submits the report. 5. The system maps the hazard to the specific route segment closest to user’s current GPS coordinate 6. The system deducts 2 points from that specific map segment's confidence score in the database, down to a clamp of -20. 7. The system adds the hazard to the “News/Hazards” screen to alert other users of the hazard.

Alternative Flows:	**AF-1 (Cancel Report):** Between step 2-4, the user selects "Cancel Report". The system dismisses the report and details filled so far and redirects the user back to his active navigation page.
Exceptions:	**EX-1 (Issue unselected):** If the user skips step 2, the system will prompt the user to select "Unwalkable", "Path blocked", "Flooding", "Something Else", "Lift down" or "Broken Surface" to confirm the severity of the hazard present and to declare the map segment has not viable.
Includes:	None
Special Requirements:	App support team may want to verify the presence of such hazards using public CCTV or site visits to avoid scams or acts of mischief by users.
Assumptions:	The user is honest and accurate in his description of the hazard and the hazard is truly severe enough to declare the map segment as inaccessible.
Notes and Issues:	None

Use Case ID:	1.6-2		
Use Case Name:	View Nearby Hazards		
Created By:	Hemanth	Last Updated By:	Hemanth
Date Created:	04/09/2026	Date Last Updated:	04/09/2026

Actor:	User
Description:	Displays a chronologically ordered list of currently reported hazards located within 5 kilometres of the user's position, so the user can be aware of unwalkable paths in their vicinity before setting out.
Preconditions:	User is authenticated and the device has supplied a current GPS location.
Postconditions:	A filtered hazard list displayed to the user. No system state is modified.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user navigates to the "News/Hazards" screen. 2. The system queries the database for all map segments have received a confirmed hazard report within the past 72 hours, or with a negative confidence score. 3. The system filters the queried hazards to those within a 5-kilometre radius of the user's current GPS location. 4. The system sorts the filtered hazards chronologically, most recently reported first. 5. The system displays the sorted hazard list to the user.
Alternative Flows:	**AF-1 (No Hazards):** No hazards exist within the 5-kilometre radius. The system displays an empty-state message confirming no hazards are currently reported nearby.

Exceptions:	**EX-1 (GPS Location Unavailable):** If a user's current location has a weak signal or the user has denied permission to his GPS location, in both cases the current user location is unavailable, no radius can be computed. The system prompts the user to enable location access or enter a location manually.
Includes:	None
Special Requirements:	Feed must load within 3.0 seconds of the user opening the News screen. Radius is strictly 5 kilometres. Sort order is descending by report time in reverse chronological order with most recent reports appearing first.
Assumptions:	Every hazard report carries a timestamp, and every map segment carries a location that can be tested against the radius.
Notes and Issues:	None